

	Statement of Work	
	Document Title: SSRL B116 & B132 BPM HVAC Replacement	
	Page 1 of 6	

Document Approval:

Approval

Originator: Andi Schoppa, SSRL Facility Steward	 Andi Schoppa
Approver: Daniel Harrington, BLS Division Director	 Daniel Harrington
Approver: Xiaobiao Huang, Accelerator Director	

2/11/2026

Revision History

Revision	Date Released	Description of Change
R0	02/02/26	Original Release

Acronyms:

- SSRL - Stanford Synchrotron Radiation Lightsource
- SPEAR - Stanford Positron Electron Accelerating Ring
- BPM – Beam Position Monitor
- SOW – Statement of Work
- BIO – Building Inspection Office
- F&O - Facilities and Operations
- ES&H – Environmental Safety & Health
- RCA - Radiologically Control Area
- GERT - General Employee Radiological training
- PAMM - Planned Access for Machine Maintenance

	Statement of Work	
	Document Title: SSRL B116 & B132 BPM HVAC Replacement	
		Page 2 of 6

1 Introduction

SLAC SSRL is seeking a qualified subcontractor to remove and replace, in its entirety, the HVAC systems that service the Beam Position Monitor (BPM) Rooms located in Buildings 116 and 132. Maintaining temperature stability is critical to the operation of the equipment located in these rooms which enables SPEAR to function. The current HVAC systems are beyond their useful life expectancy and require replacement.

Because the BPM's are essential to accelerator operations, all construction activities will need to occur while the SPEAR accelerator is shut down. The shut down period that provides sufficient time to perform this work is during SSRL's annual summer shutdown. The 2026 Summer Shut Down for construction activities is scheduled between July 21, 2026 through October 1, 2026. SSRL intends to perform as much of the work associated with this project as practical during the Summer 2026 Down.

An operational HVAC system is required by September 19, 2026. If equipment purchasing lead times impact the ability of the project to achieve operational HVAC systems by that date, the project shall be phased to complete as much work as possible during the Summer 2026 down, with the remaining work occurring during the Summer 2027 shut down.

SSRL is a Scientific User Facility comprised of an electron accelerator and storage ring, radiation shielding structures, beam line hardware, laboratories and support equipment, all of which house sensitive scientific equipment. Care must be taken to protect and minimize impact to these areas during construction.

All the provisions of the Contract Documents, including without limitation the General Requirements, drawings and specifications shall apply to the work described in this Statement of Work (SOW).

2 Project Scope of Work

The subcontractor shall include all oversight, labor, tools, materials, equipment, and other services needed to safely complete the work outlined in this SOW and in accordance with the attached BIO approved documents BIO PRS #25-253 & 25-254 to deliver fully functional HVAC systems for both BPM rooms. The drawings and specification sets are identical for each permit but were submitted under two different PRS numbers to delineate each location for inspection purposes, as required by BIO. Work shall be executed with minimal disruption in full compliance with SLAC, DOE, and ES&H standards.

The Project Scope includes but is not limited to the following:

1. Provide submittals for all materials and equipment to be used by the subcontractor for review by the Engineer of Record (EOR) and SLAC, including but not limited to fan coil units, condensers, fire smoke damper/detectors, refrigerant piping, electrical components, ductwork, and equipment supports, concrete pad materials, and bollards.
2. Perform a field survey of the work area to confirm all dimensions, routing and materials necessary for a complete assembly. Certain areas may only be accessed while the SPEAR Accelerator is off during a PAMM or shutdown period. Refer to the attached SSRL Operating Schedule 2025-2026. Coordinate access with the SSRL Building Manager.
3. Mobilize, provide temporary facilities, dust control and other environmental controls. Protect all surfaces during demolition and construction activities.

4. Coordinate with SLAC for utility scanning in demolition areas to identify existing utilities or obstructions prior to work. Potholing may be required to minimize damage to underground utilities adjacent to work zones and should be confined to the new housekeeping pad locations.
5. Demolish all existing HVAC equipment, piping, conduits, and supports not designated for reuse and transport materials to an on-site salvage or disposal location per Waste Management direction. Coordinate with Facilities & Operations (F&O) and the Building Manager for proper drainage and disposal of hazardous and non-hazardous materials, including legacy refrigerants.
6. Demolish existing asphalt as required to install the new concrete housekeeping pads and bollards for the condenser units per drawings. An excavation permit will be required for the bollard installation due to the excavation depth.
7. Layout pad and bollard locations to minimize impacts to the infield parking while maintaining required equipment working clearances. Mark layouts on the pavement for SLAC approval prior to construction.
8. Coordinate with SLAC for all code required BIO and Special Inspections for the new pad and bollards for rebar, concrete, and equipment anchoring.
9. Furnish and install complete new Vertiv Liebert HVAC equipment and accessories necessary to provide complete and operational systems.
10. Install new ductwork, supports, and ceiling mounted supply registers with mounting frames per the drawings. Coordinate the routing to avoid lights, conduits, fire sprinklers piping, and other ceilings obstructions.
11. Furnish and install new refrigeration piping, supports, and insulation between the fan coil units and new condenser unit locations per the drawings. Charge systems in accordance with the specified refrigerant requirements.
12. Provide manual volume balancing dampers as shown. Balance airflow and temperature per the drawings.
13. Provide a deferred fire alarm submittal package, including drawings, calculations (if required), design and testing criteria for smoke detector and dampers tie-ins for automatic HVAC shutdown. Submittal for review by SLAC BIO and the Fire Marshall.
14. Provide condensate lines as shown on the drawings.
15. Provide duct insulation for both the supply and return ductwork.
16. Fire-seal all penetrations through any fire walls or barriers and weather-seal non-fire rated walls to make airtight construction.
17. Patch and repair any demolished or damaged drywall to match existing surfaces.
18. Provide new electrical infrastructure including conduits, feeders, breakers, switches, grounding, disconnects and maintenance receptacles as shown on the drawings. Ensure locations and layout comply with code required clearances.
19. All electrical work shall be performed by California-certified journeyman electricians in accordance with 29 CFR 1910 Subpart S, Electrical, NFPA 70E, Standard for Electrical Safety in the Workplace Section 110.2
20. Consider all drawings and all sections of this specification as a whole and provide all work shown or reasonably implied.

 NATIONAL ACCELERATOR LABORATORY	Statement of Work	
	Document Title: SSRL B116 & B132 BPM HVAC Replacement	
		Page 4 of 6

21. Recognize that absolute accuracy of the drawings cannot be guaranteed. Verify field conditions and report discrepancies to the SSRL PM or Building Manager. Unauthorized changes shall be at the Contractor's risk and expense.
22. With prior SLAC approval, modify equipment locations as necessary to accommodate field conditions. Clearly document all changes on shop and coordination drawings and submit to SLAC for approval prior to ordering or fabrication. Final as-built documentation will be required at the end of the project to receive final payment.
23. The Contractor shall provide all necessary offsets, transitions, and modifications required to complete the work at no additional cost.
24. Follow manufacturer installation instructions unless otherwise noted. Resolve conflicts prior to commencing work.

All safety and technical submittals shall be submitted to SLAC for review and approval in accordance with the subcontract requirements and project schedule as defined in this Statement of Work. SLAC will issue a formal Notice to Proceed (NTP) before the subcontractor can perform work on site.

The SSRL Complex, where this work will take place is a designated Radiation Controlled Area. Workers are required to complete GERT Training in addition to Subcontractor Worker Safety Training as required by SLAC.

3 Deliverables

- Equipment and material submittals demonstrating compliance with the plans and specifications.
- Detailed schedule with durations for submittals, equipment procurements, construction activities and phasing plan if necessary.
- Deferred BIO Fire Alarm Submittal for smoke detector tie-ins.
- Operation and Maintenance documentation at project closeout
- As-built drawings at project closeout

4 ATTACHMENTS

1. BIO PRS #25-253 B116 SSRL BPM Room HVAC Replacement Documents
2. BIO PRS #25-132 B132 SSRL BPM HVAC Replacement Documents
3. SSRL SPEAR3 Operating Schedule 2025-2026
4. SLAC BIO Adopted Codes and Standards (Updated 01/01/2025)

5. BUILDING INSPECTION OFFICE CRITERIA

The Contractor shall comply with all Building Inspection Office Adopted Codes and Standards as identified in the attached reference documents. The Contractor is required to follow the BIO Review and Authorization Manual which outlines requirements for plan submission and review,

inspections, and project closeout.

6. TEMPORARY FACILITIES; CONSTRUCTION POWER; PORTABLES and LAYDOWN AREAS

A. The work of this proposal package shall include all labor, material, tools, equipment, hoists, scaffolding and services needed to perform the demolition and construction work, and installation described in the specifications, or that may be reasonably inferred as being necessary for the completion and serviceability for the intended use of all items of work.

1. Construction shall meet or exceeds the requirements set forth in the approved drawings and specifications.
 - 1.1. Construction Limits: Site signage, traffic control, laydown areas, and location of temporary toilets/facilities shall be approved by SLAC. Existing driveways and fire lanes affected by construction shall remain open at all times during construction of the facility. A minimum of one-way traffic must be maintained in areas where construction obstructs roadways and pathways. On-site fire hydrants and Building perimeter areas shall be accessible by fire trucks as required by the SLAC Fire Marshal's Office.
 - 1.2. Temporary Construction Water and Power: Subcontractor to coordinate with SLAC's Construction Manager (CM) for temporary power and construction water locations.
 - 1.3. Environmental, Safety, Health and Quality (ESH&Q) Criteria: Consideration for the safety and health of the employees and public, and mitigation of impacts to the environment shall take precedence over other aspects, such as function, cost, and expediency. Contractor to adhere and follow SLAC's ESH Manual.
 - 1.3a Building Inspection Office Criteria: The Contractor shall comply with all Building Inspection Office (BIO) adopted codes and standards (see attached reference documents). In addition, the Contractor is required to follow the BIO Review and Authorization Manual, which outlines requirements for plan submission and review, inspections, and project closeout.
 - 1.4. Existing Conditions: Contractor shall protect all existing material and equipment not scheduled for removal or modification.
 - 1.5 Working in an Operating Building: The GC shall make every effort to minimize utility outages affecting adjacent operating labs, all outages shall be coordinated with the SLAC CM and PM with at least 2-week notice. Disruptive work shall be clearly identified in the GC's 3 week look ahead schedule.

7. CONSTRUCTION PACKAGE SCHEDULE and MILESTONES

Work on site is anticipated to occur during SSRL's Summer Shutdown period, scheduled from July 21, 2026 through September 19, 2026 for construction activities.

Description	Tentative Date (Subject to Change)
Proposal due	2 weeks after the bid walk site visit

	Statement of Work	
	Document Title: SSRL B116 & B132 BPM HVAC Replacement	
		Page 6 of 6

Contract Award	2 weeks after proposal
Submittals Schedule	1 week after award
NTP	1 week after award
Project Execution – Work on site	July 21 thru Sept 19, 2026 & 2027 Summer Down TBD if needed

If access is needed to review areas that are located in exclusion zones prior to the summer down, refer to the attached SPEAR3 Operating Schedule 2025-2026 for PAMM days. SSRL schedules a PAMM day each month to shut down the accelerator to allow for maintenance activities. Coordinate with the SSRL Facility Steward or Building Manager at least a week in advance to enable time to process access requests.